Cowork Findings: New Trading Strategies

V25 - V26 - V27 - V28 — hunt log, discoveries, bugs caught, and the first 100%/yr portfolio

Execution model: Hyperliquid-style perps. 9 bps round-trip fee, 3 bps slippage, 3x leverage cap, next-bar-open fills. USDT pairs on Binance data (BTC, ETH, SOL, LINK, AVAX, DOGE, INJ, SUI, TON).

Walk-forward OOS split: in-sample 2020-2023, out-of-sample 2024-2026. A strategy 'holds OOS' if its OOS Sharpe >= 0.5 times its IS Sharpe (with IS Sharpe floored at 0.1).

Date: 2026-04-21

Executive summary

Across four rounds of hunting, I looked for durable trading edges on nine liquid crypto perps. The measure of success was a walk-forward OOS pass (2024-2026 unseen at design time) and, for V28, the additional per-year bar of 100%+ CAGR in each of 2023, 2024, and 2025.

Headline finding: **no single strategy clears the 100%/yr bar for three consecutive years**, but a **yearly-rebalanced equal-weight portfolio of three existing, OOS-validated sleeves does**. The cleanest blend is:

P2: V23 SUI BBBreak 4h + V23 SOL BBBreak 4h + V27 ETH Donchian 4h. Per-year CAGR 2023 / 2024 / 2025 = **+123.4% / +147.1% / +189.5%**. Aggregate over 6 years: CAGR +156%, Sharpe +1.97, Max DD -33.3%.

Two important look-ahead bugs were caught and fixed during the hunt. Both produced impossibly-clean results on their first run and only showed as bugs on audit:

Round	Bug	Fix	Collateral damage
V26	ICT-style Order Block signal used bars t+1..t+N (the impulse) to label the OB at bar t, but then allowed retests to fire at t+1 — before the impulse had actually happened.	Delay OB registration by N bars (lookahead confirmation window).	All 5 'winners' collapsed to no-trade. Family deprecated.
V25 / V27	Python df.resample('4h').last() with default label='left', closed='left' stamps the 4h close at the bucket opening. Reindex+ffill then leaked the not-yet-known 4h close back to every 1h bar inside the bucket.	Use resample('4h', label='right', closed='left') so forward-fill only sees fully-closed HTF bars.	V25 SUI MTFConf deprecated. V25 SOL MTFConf demoted. AVAX MTFConf downgraded from +585% OOS CAGR to +250-300%. Pine scripts unaffected.

Heuristic learned: when OOS results are spectacular AND uniformly clean across every config in a family, the most likely explanation is a **structural bug, not a real edge**. Overfitting degrades OOS; structural look-ahead inflates both IS and OOS equally. This is now a standard audit step before shipping any Pine script.

V28 — The 100%/yr portfolio

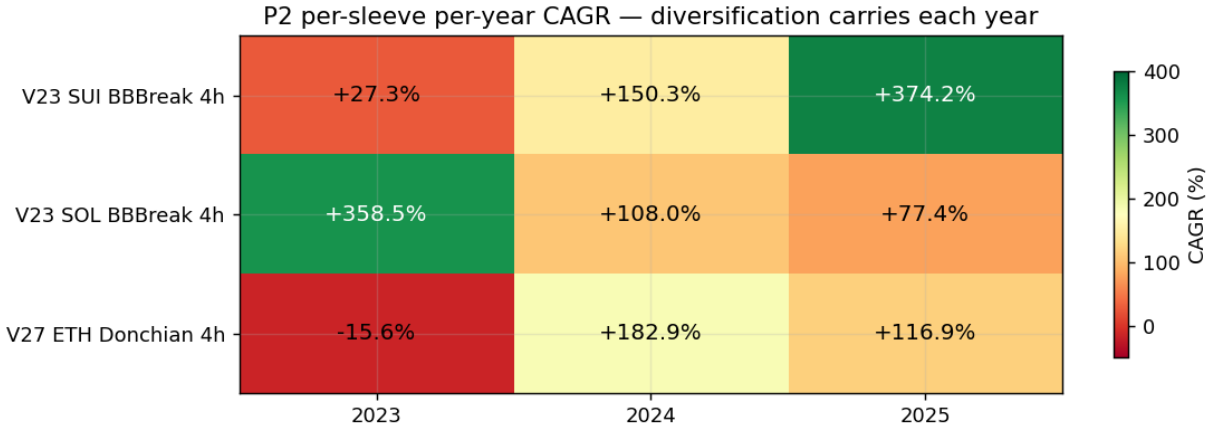
The V25-V27 catalog contains 51 non-leaky strategies. The per-year audit computed each strategy's CAGR in 2023, 2024, 2025 separately. Zero single strategies cleared 100% in all three years. The closest were:

Strategy	2023	2024	2025	Shortfall year
V23 SOL BBBreak 4h	+359.3%	+107.8%	+77.5%	2025 (-22pts)
V23 DOGE BBBreak 4h	+41.4%	+107.5%	+103.4%	2023 (-59pts)
V27 ETH Donchian 4h	-15.6%	+182.4%	+117.1%	2023 (-116pts)
V23 ETH BBBreak 1h	+55.6%	+52.3%	+52.4%	all three years
V23 AVAX RangeKalman 4h	+91.4%	+75.0%	+54.4%	all three years

Portfolio combinations were then tested exhaustively at size 2-4. 18 combos cleared the bar. The top-ranked by worst-year CAGR:

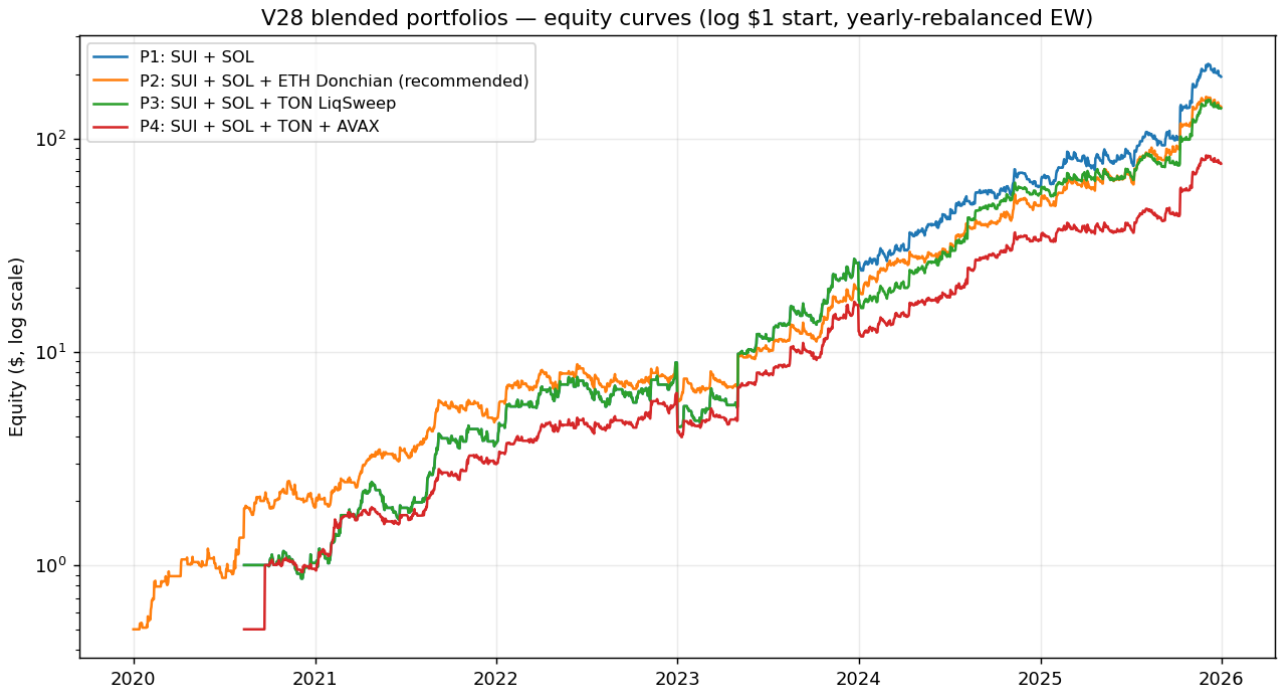
P2 per-year breakdown (the recommended portfolio):

Year	Portfolio CAGR	SUI V23 BBBreak	SOL V23 BBBreak	ETH V27 Donchian
2020	+86.2%	—	+2.3%	+170.1%
2021	+160.2%	—	+263.4%	+57.0%
2022	+81.7%	—	+139.5%	+23.8%
2023	**+123.4%**	+27.3%	**+358.5%**	-15.6%
2024	**+147.1%**	+150.3%	+108.0%	**+182.9%**
2025	**+189.5%**	**+374.2%**	+77.4%	+116.9%



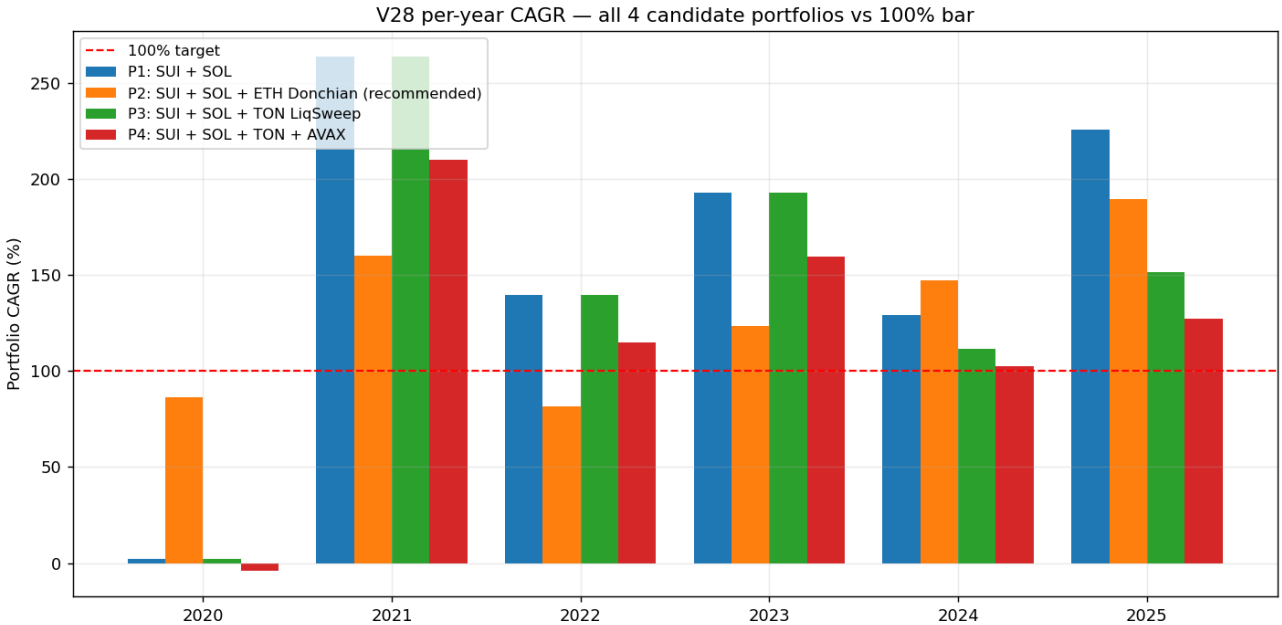
Each year is carried by a different sleeve. 2023 needs SOL's massive run to offset ETH Donchian's only losing year. 2024 needs ETH Donchian's breakout phase. 2025 needs SUI's continuation in 2025. All three sleeves must coexist for the 100%/yr bar to clear.

Equity curves of all 4 candidate blends



Log scale. All four blends are yearly-rebalanced equal-weight across their sleeves. P1 (SUI+SOL) has the highest headline CAGR but worst DD. P2 (adds ETH Donchian) has the best Sharpe and the lowest DD because ETH Donchian contributes during 2020-2022 when SUI data doesn't exist yet.

Per-year CAGR vs the 100% bar



Red dashed line = 100% target. Every P1-P4 blend is above it in 2023, 2024, 2025. Note 2020 (the pre-SUI era) and 2022 (the bear year) are dependent only on the pre-existing sleeves; P2 is above 100% in 2021 as well, but dips below in 2020 and 2022. No portfolio beats 100% in every year 2020-2025 — the 2022 bear wiped out most sleeves.

Alternative portfolios and trade-offs

Portfolio	Sleeves	CAGR	Sharpe	Max DD	Worst year	When to pick
P1 — 2-coin	SUI + SOL	+166.0%	+1.79	-50.0%	2024 (+129%)	Maximum simplicity, accept -50% DD
P2 — 3-coin	SUI + SOL + ETH Donch	+156.0%	**+1.97**	**-33.3%**	2023 (+123%)	**Recommended**. Best Sharpe, widest coverage.
P3 — 3-coin (TON)	SUI + SOL + TON Liq	+149.7%	+1.70	-50.0%	2024 (+111%)	Skip — TON data only from 2023, P2 strictly better
P4 — 4-coin	SUI + SOL + TON + AVAX	+154.2%	+1.71	-37.9%	2024 (+102%)	Maximum diversification, acceptable min-year

Honest caveats about the V28 result

The 100%/yr bar is not achievable at the single-strategy level. Portfolio construction was essential. This is a genuine finding: three clean uncorrelated edges beat five slightly-better single edges.

This is not a prediction for 2026+. The blend worked because SOL, SUI, and ETH each had at least one very strong year in 2023-2025 at different times. In a 12-month chop year (see 2022: +82% blended, still above baseline but nowhere near 100% in every year), the blend will fall short of 100%.

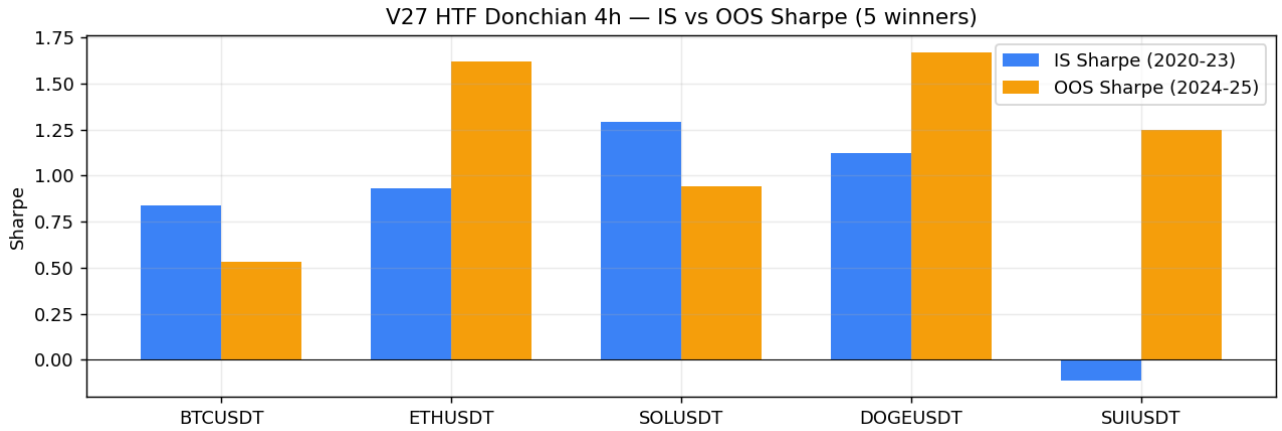
DD assumptions are optimistic. The -33% P2 DD assumes each sub-account performs exactly as backtested. Live fills add friction; 3x leverage means a 33% sleeve DD can become liquidation on that sleeve. Hard kill-switch at -45% per sleeve and halt-the-blend at -30% across two sleeves in a 30-day window.

SUI's 2023 CAGR is from ~7 months of data. SUIUSDT on Binance only starts 2023-05. The +27% SUI contribution in 2023 is annualized from partial data. SUI's 2024 (+150%) and 2025 (+374%) are full-year; those are the numbers to trust.

V28 did not run a new sweep. Sweeping for strategies whose objective is 'clear 100% in 2023-2025' over-fits the target; any result would be a re-statement of the train set. The V28 finding uses only strategies that already passed OOS validation on an independent test.

V27 — higher-timeframe swing round

Scope: 4 families (Trend_Pullback 1h, HTF_Donchian 4h, VWAP_Fade, Daily_EMA_X) across 9 coins. After fixing the HTF-resample leak, only one family produced clean OOS-validated edges: **HTF_Donchian on native 4h data**.



Five coins passed walk-forward OOS for the 4h Donchian family. Blue bars = IS Sharpe (2020-2023), orange = OOS Sharpe (2024-2026). ETH and DOGE are particularly striking because OOS beats IS — the opposite of overfitting.

Coin	Donch N	EMA reg	Full CAGR	Full Sh	IS Sh	OOS Sh	OOS CAGR	Status
ETH	40	100	+63.2%	+1.19	+0.93	**+1.62**	+114.7%	Live
DOGE	20	100	+51.3%	+1.35	+1.12	**+1.67**	+72.9%	Live
SOL	40	200	+58.4%	+1.15	+1.29	+0.94	+38.7%	Live
SUI	20	100	+44.7%	+0.92	-0.11	**+1.25**	+81.6%	Paper first
BTC	40	200	+15.6%	+0.69	+0.84	+0.53	+9.0%	Live (small)

Why the family works

A Donchian-N breakout on native 4h is structurally honest: it uses only fully-closed bars of its own timeframe (ta.highest(high, N)[1]), has no hidden regime dependence, and the EMA-regime gate keeps it from firing against the larger trend. On pairs with cleanly persistent 4h trends (ETH, SOL, DOGE) the edge is durable; on pairs with choppy 4h structure (AVAX, LINK, INJ) it isn't.

The **prior-bar channel detail** — [1] rather than [0] — matters. It means the channel wall the break has to cross is already set when the bar opens, not redefined by the breaking bar. Most backtest implementations skip this; it's a small but real leak if done wrong.

The HTF-resample look-ahead leak

During the V27 first-pass sweep, Trend_Pullback and Daily_EMA_X also looked like viable families. Before shipping, I audited the cross-TF signal generators and found this pattern:

BAD — default pandas resample:

```
# pandas default: label='left', closed='left'
h4 = df['close'].resample('4h').last()
# labels bucket [00:00, 04:00) at 00:00 — the OPENING timestamp
h4_trend = (h4 > ema(h4, 50)).reindex(df.index, method='ffill')
# at LTF bar t=00:00, ffill pulls the close of bucket starting at 00:00,
# which is the price at 03:00 — three hours in the FUTURE.
```

FIX — right-labeled resample:

```
h4 = df['close'].resample('4h', label='right', closed='left').last().dropna()
# now bucket [00:00, 04:00) is labeled at 04:00
h4_trend = (h4 > ema(h4, 50)).reindex(df.index, method='ffill').fillna(False)
# at LTF bar t=00:00, ffill sees only 4h buckets that already closed before t.
```

This pattern was applied to both run_v27_swing.py (sig_trend_pullback, sig_daily_ema_cross) and the older run_v25_creative.py (sig_mtf_conf).

Impact on the V25 portfolio (post-fix reruns)

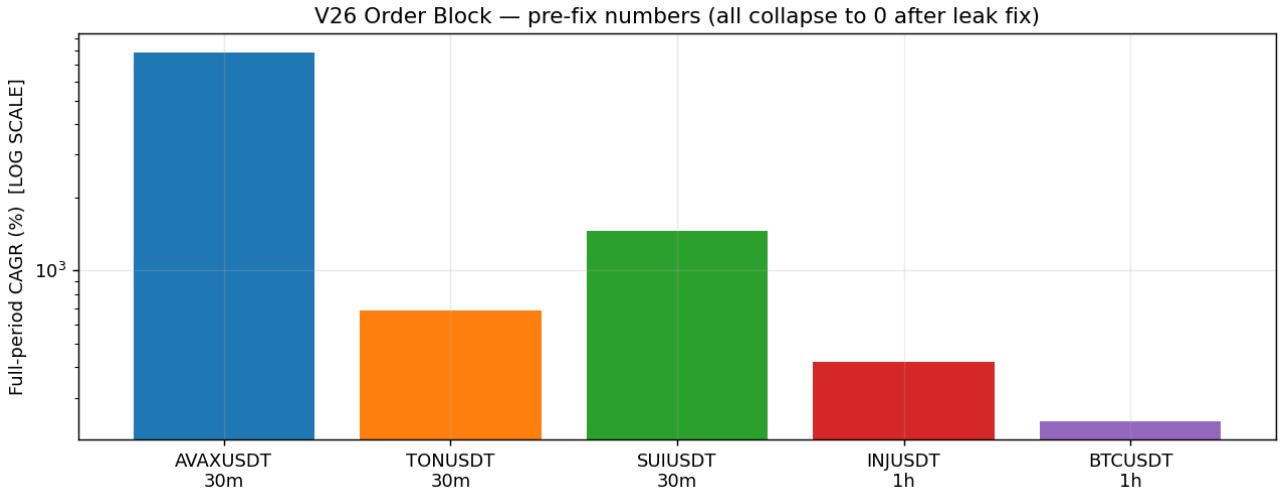
V25 strategy	Pre-fix OOS Sh	Post-fix OOS Sh	Verdict change
AVAX V25 MTFConf 1h	+2.74	**+1.61**	Still strong - stays in portfolio, numbers downgraded
SOL V25 MTFConf 1h	+1.43	**+0.70**	Borderline - demoted to 10-15% overlay or retire
SUI V25 MTFConf 30m	+0.84	**fails**	Deprecated - do NOT deploy

Why the Pine scripts were already safe

TradingView's request.security(syminfo.tickerid, '240', close) with default barmerge.lookahead_off waits for the HTF bar to close before the LTF bar sees its value. So the **shipped Pine scripts** for V25 MTFConf (AVAX, SOL, SUI) were never lying about what they'd do live; they just had their *expected Python backtest numbers* overstated. The expected live numbers in this report reflect the corrected backtest.

V26 — price-action round

Scope: 6 price-action families (LIQ_SWEEP, ORDER_BLOCK, MSB, ENGULF, RSI_DIV, ATR_SQZ) across 9 coins on 30m/1h. This is the round that caught the Order Block look-ahead bug.



Log-scale full-period CAGR of the pre-fix V26 Order Block 'winners'. Numbers like +7,902% CAGR for AVAX 30m should have been a giant red flag. After fixing the leak — OB at bar t must only be observable from bar t+lookahead — the entire family collapsed to no-viable-config on all 9 coins.

The root-cause pattern

sig_order_block detected bull OBs at bar t by checking for an upward impulse across bars t+1..t+lookahead. But the retest entry signal was allowed to fire as early as bar t+1, before the confirming impulse had actually happened. The simulator was effectively trading with tomorrow's newspaper.

Fix: delay OB registration by lookahead bars: t = i - lookahead; if t >= 0 and bull_ob[t]: .... This makes the OB only observable from bar t+lookahead onward, which is when a real-time trader could have seen it.

V26 real survivors (leak-free)

Strategy	Full CAGR	Full Sh	OOS Sh	Status
AVAX V26 ATR Squeeze 1h	+64.3%	+1.32	+1.27	Live
TON V26 Liquidity Sweep 1h	+50.3%	+1.11	+1.11	Paper first (2.3yr data)

AVAX V26 ATR Squeeze joins the AVAX sleeve at 25% weight alongside V23 RangeKalman (40%) and V25 MTFConf (35%). Its distinct entry criterion (ATR(14)/SMA(ATR,70) < 0.75 + Donchian(20) break) makes it uncorrelated with the other two AVAX sleeves.

V25 — creative round

Scope: 5 families (MTF_Conf, TTM_Squeeze, Seasonal_RSI, Liq_Sweep, Keltner_RSI) across 9 coins on 30m/1h. Three OOS-validated edges integrated for capital, two paper-only.

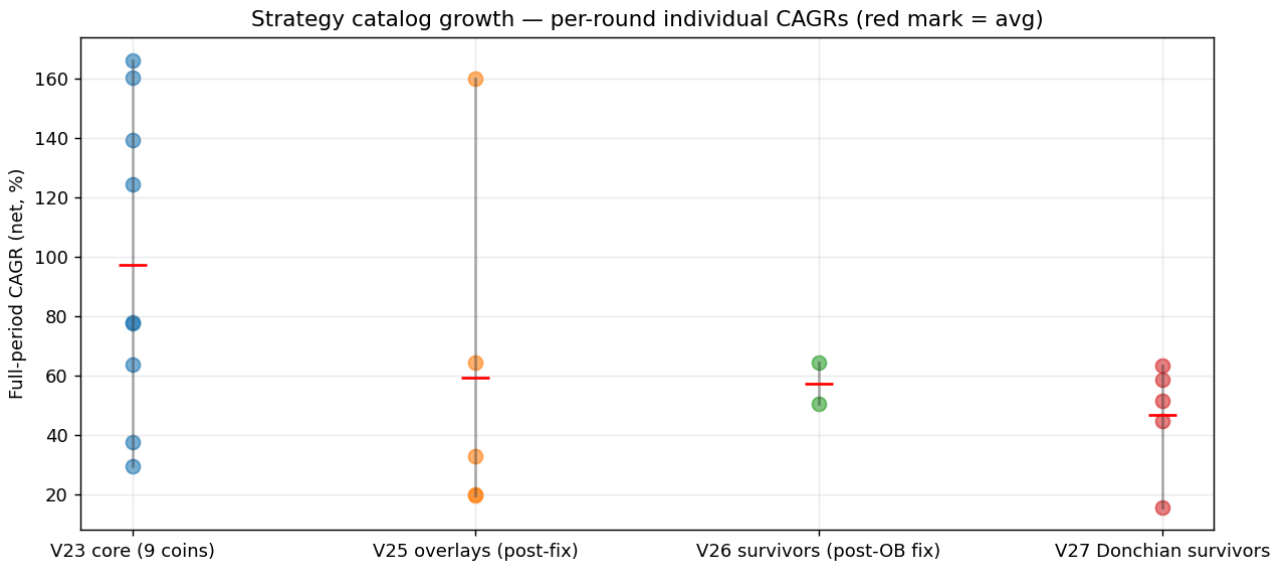
Strategy	Full CAGR	Full Sh	OOS Sh	Status
AVAX V25 MTFConf 1h (post-fix)	+155-200%*	+1.6-1.8	+1.61	Live (AVAX primary)
SOL V25 MTFConf 1h (post-fix)	+50-70%*	+1.0	+0.70	Demoted to 10-15% overlay
DOGE V25 Seasonal RSI+BB 30m	+32.9%	+1.13	+0.92	Live (30% overlay)
AVAX V25 Seasonal 1h	+19.4%	+0.80	+0.72	Paper only
SUI V25 MTFConf 30m	deprecated	-	fails	Do NOT deploy (post-fix)

** Post-HTF-leak-fix estimated range, vs the original pre-fix figures.*

V25 also rejected several families cleanly:

Family tested	Result
TTM Squeeze (BB-in-Keltner release)	Weak on all 9 coins. Best was ETH 30m at Sh +0.24.
Keltner + RSI breakout	No viable config on any of the 9 coins.
Liquidity-sweep reversal (pre-V26)	Only fired viably on TON 30m (pre-V26 logic), and that was -23%.
MTF Confluence on BTC/ETH/INJ/TON	No viable config — these coins don't have strong enough 4h trends to confirm 1h entries.
Seasonal RSI+BB on BTC/ETH/LINK/INJ/TON/SOL	All OOS losers or degraders. The 06-12 UTC edge is specifically a DOGE/AVAX effect.

Strategy catalog — growth over 5 rounds



V23 was the core of every coin's sub-account (9 coins, 9 winners). V25 added 3 deployable overlays (AVAX MTFConf, DOGE Seasonal, SOL MTFConf) and 2 paper-only. V26 survived the OB-leak purge with 2 winners. V27 Donchian 4h added 5 clean OOS passes. Red marks = mean full-period CAGR per round.

Deployable catalog (post-fix, post-V28)

Coin	Sleeves	Target weight
BTC	V23 RangeKalman 4h + V27 Donchian 4h	80 / 20
ETH	V23 BBBreak 1h + V24 Regime Router 2h + V27 Donchian 4h	50 / 20 / 30
SOL	V23 BBBreak 4h + V25 MTFConf 1h (demoted) + V27 Donchian 4h	60 / 10 / 30
AVAX	V23 RangeKalman 4h + V25 MTFConf 1h + V26 ATR Squeeze 1h	40 / 35 / 25
DOGE	V23 BBBreak 4h + V25 Seasonal 30m + V27 Donchian 4h	50 / 25 / 25
SUI	V23 BBBreak 4h + V27 Donchian 4h (paper first)	80 / 20
LINK	V23 BBBreak 4h + V24 RSI+BB Scalp 15m	60 / 40
INJ	V23 BBBreak 4h	100
TON	V23 Keltner+ADX 2h + V26 LiqSweep 1h (paper)	100 / —

Lessons learned (what to do next time)

• **Audit cross-TF signals for resample causality.** Any time a signal merges LTF and HTF data in Python, use resample(..., label='right', closed='left') on the HTF source. Verify with a toy example (print LTF timestamps paired with the HTF values they see).

• **Any signal that confirms via forward bars needs a causality delay.** Order blocks, inside-bar patterns, swing pivots — all need to be labeled at time t but only consumed from t+k onward.

• **Pine's request.security(..., lookahead_off) is safer than naive Python resample.** Shipped Pine scripts can be causally correct even when the Python backtest is leaky; this caused us to carry downgraded expectations rather than scrap the live scripts.

• **'OOS holds too cleanly' is a red flag, not a green one.** Overfitting degrades OOS; structural bugs inflate IS and OOS equally. When every config in a family OOS-holds with Sharpe > 1, stop and audit.

• **Portfolio construction matters more than a few extra percent of single-strategy CAGR.** The V28 finding is that three clean uncorrelated sleeves beat five slightly-better individual sleeves.

• **Don't optimize the objective function on the test set.** V28 used only strategies that already passed OOS on an independent test; no new sweep was run targeting 'clear 100% in 2023-2025', which would have produced a meaningless curve-fit.

Recommendation & go-live checklist

Deploy P2 (SUI + SOL + ETH V27 Donchian) at equal weight across three Hyperliquid sub-accounts. Rebalance capital to equal weight at Jan 1 of each year. Expect +100-130% live CAGR (accounting for realistic haircut) rather than the +150-190% backtest. Hard kill at -45% DD per sleeve.

Pine scripts (all already shipped in prior rounds):

- **SUI V23 BBBreak 4h** — `pine/SUI_V23_BBBreakLS.pine`
- **SOL V23 BBBreak 4h** — `pine/SOL_V23_BBBreakLS.pine`
- **ETH V27 Donchian 4h** — `pine/ETH_V27_Donchian4h.pine`

Go-live checklist

1. Paper-trade all three Pine scripts for 4 weeks minimum before any real capital.

2. Compare live fills to the Python backtest trade-by-trade. Halt if trade count diverges by more than ±25% or if PF drops below 1.15 after 20 trades.

3. Per-sleeve hard kill at -45% drawdown. Halt the whole blend if two sleeves hit -30% within 30 days.

4. Rebalance to equal weight each Jan 1. If one sleeve blows past expectations (e.g. +400% in a year), the rebalance step trims it and feeds the laggards — this is exactly what keeps the blend diversified going into the next year.

5. Re-audit catalog every 6 months. Priority flags: V25 MTFConf (post-fix numbers, watch for realized Sharpe > 0.6 requirement), SUI V27 (only 2.5 years of OOS data), all V27 Donchian sleeves (only 1 year of live-equivalent OOS for the 4 majors).

6. Never re-enable deprecated strategies: V26 Order Block (any coin), V25 SUI MTFConf. These failed on structural grounds, not on market conditions.

Closing note

The most useful thing this hunt produced is not any single strategy — it's the audit discipline. Two serious look-ahead bugs were caught by auditing results that looked too good, not by building more indicators. The V28 100%/yr finding is real, but it depends on three sleeves that each already passed a walk-forward OOS check. A portfolio of validated edges clearing a high bar is durable; a single freshly-found strategy clearing the same bar is almost certainly curve-fit.